The OneBitHHL quantum filter for segment-level track finding in a planar LHCb VELO toy: algorithm, padding, spectral kernel, and the $\gamma = 3$ rejection notch

Internal report — Quantum Track Reconstruction working note
Nikhef / Vrije Universiteit, Amsterdam

May 27, 2026

Abstract

We document the one-bit phase-estimation linear-solver variant (OneBitHHL / OneBQF) used for segment-level track finding in the planar LHCb VELO toy `lhcb_velo_toy` (v2.0.0). The Hopfield-style segment Hamiltonian, the cost function, the choice of ε, γ, δ, and the segment activation threshold τ are stated explicitly. The action of OneBitHHL is derived as a *spectral filter* $f(\lambda) = \cos(\pi\lambda/(2c))$ with $c = \gamma + \delta$; this is shown to differ qualitatively from the matrix inverse delivered by HHL. With the nominal classical operating point ($\gamma = 3$, $\delta = 1$, $\varepsilon = 2\,\text{mrad}$, $\tau = 0.35$, $\sigma_{\text{res}} = 5\,\mu\text{m}$, $\sigma_{\text{scatt}} = 0.1\,\text{mrad}$, 1% random hit drop), the spectrum of A straddles the rejection notch at QPE phase $\varphi = 0.5$ and one quarter of the eigenmodes are completely wiped. At low track multiplicities ($T \leq 10$) this manifests as a $\text{eff}_Q \approx 0.73$ efficiency floor; at high multiplicities ($T \geq 200$) the same kernel inflates the false-segment activation count by up to a factor four, producing a runaway false rate even at exact-statevector readout. A post-processing threshold sweep shows that raising τ from 0.35 to 0.40 suppresses the high-T false rate from $\approx 55\%$ to $\approx 1\%$ at $T = 500$ and from $\approx 83\%$ to $\approx 0\%$ at $T = 1000$, at a 0.5–4% efficiency cost. Statevector (E_1) and finite-shot sampling (E_2) readouts agree within 1% at every T in the sweep, ruling out shot noise as the failure mode.

1 Introduction

The `lhcb_velo_toy` package generates planar VELO events and reconstructs tracks via two equivalent routes: an inner triplet-cone segment graph passed to either a sparse linear solver (the "classical" route) or to a quantum linear-system algorithm. This note characterises the second route and stands alongside the classical-only segment-level report [1]. A companion document compares the two solvers head-to-head; here we restrict ourselves to defining the algorithm, identifying its failure modes, and demonstrating that they are not statistical artefacts.

2 Event model and reconstructable-track definitions

Geometry. Five planar modules at $z \in \{33, 66, 99, 132, 165\}$ mm; half-width 40 mm; PV uncertainty $\sigma_{PV} = (0, 0, 1)$ mm; opening cone $\phi_{\max} = \theta_{\max} = 0.2$ rad; particle: pion.

Detector effects. Hit-position resolution $\sigma_{\text{res}} = 5\,\mu\text{m}$; multiple-scattering angle $\sigma_{\text{scatt}} = 10^{-4}$ rad; random hit drop $p_{\text{drop}} = 0.01$.

Reconstructable track. At least three reconstructable hits remaining after the drop; matched-hit purity ≥ 0.70; minimum hit efficiency 0 (LHCb default).

3 Segment graph and the Hopfield Hamiltonian

Triplets of hits (h_a, h_b, h_c) with h_a, h_b, h_c on consecutive modules are collected; the central hit h_b defines a *segment node* whose direction is the unit vector along $\overrightarrow{h_b - h_a}$ stitched with $\overrightarrow{h_c - h_b}$. Two segment nodes i, j sharing the central hit h_b are connected by an edge if the angle between their direction vectors is below the tolerance ε, equivalently $\cos\theta_{ij} > \cos\varepsilon$. The total number of segment nodes scales empirically as $N_s \sim T^p$ with the fitted exponent p stable at 3.93 for $T \in [2, 5]$ falling to 3.91 at $T = 1000$ [1].

The activation amplitude $s \in \mathbb{R}^{N_s}$ minimises

$$\mathcal{H}(s) \;=\; \tfrac{1}{2} s^\top A\, s \;-\; b^\top s, \qquad A = (\gamma + \delta)\mathbb{1} - W, \qquad b = \delta \cdot \mathbb{1}_{N_s}, \tag{1}$$

with $W \in \{0,1\}^{N_s \times N_s}$, $W_{ij} = 1$ if and only if segments i, j share a hit and pass the ε alignment cut [2]. Symmetry is enforced by $W \leftarrow W + W^\top$ and the diagonal set to 0. Because A is real symmetric with strictly positive diagonal $(\gamma+\delta)$ and unit off-diagonals limited to the shared-hit neighbourhood, A is SPD for the parameter choices considered here. The classical solver returns $s^* = A^{-1}b$; the quantum solver of Section 4 returns a spectrally filtered approximation.

Parameter choices. The base values used throughout this report are

$$\gamma = 3, \quad \delta = 1, \quad \varepsilon = 2\,\text{mrad (fixed override)}, \quad \tau = 0.35. \tag{2}$$

The values of γ, δ, τ are the §14e operating point of `segment_level_analysis.ipynb` [3]. The ε override (replacing the formula-driven $\varepsilon \sim 2\sigma_{\text{scatt}}$) is deliberate and applied identically on the classical and quantum sides for direct comparability.

The two Hopfield fixed-points relevant for the activation threshold are the false-segment attractor $\delta/(\gamma + \delta) = 0.25$ and the outer-truth attractor at 0.36; the choice $\tau = 0.35$ sits between them.

4 The OneBitHHL spectral filter

4.1 Padding and Hamiltonian simulation

The pair (A, b) is padded to dimension $N = 2^n$, $n = \lceil \log_2 N_s \rceil$; the right-hand side is padded with ones, then normalised, $|b\rangle = b_{\text{pad}}/\|b_{\text{pad}}\|_2$. Provided the diagonal is strictly constant $c = \gamma + \delta$ (true by construction for the unmodified SimpleHamiltonianFast), the time-evolution factorises $e^{-iAt} = e^{-ict}e^{+iBt}$ with $B = c\mathbb{1} - A$ off-diagonal. Each unit off-diagonal entry is implemented as a multi-controlled RX$(2t)$ Givens rotation; with the choice

$$t \;=\; \pi/c \;=\; \pi/(\gamma + \delta), \tag{3}$$

the evolution is exact (no Trotter step).

4.2 Phase estimation, X–CX–X gadget, and the cosine kernel

With one time qubit, QPE produces the entangled state

$$|\Psi\rangle \;=\; \sum_k \beta_k \big[\, \cos(\pi\varphi_k)\; |0\rangle_t \;-\; i\sin(\pi\varphi_k)\; |1\rangle_t \,\big]\, |\lambda_k\rangle\,, \quad \varphi_k = \lambda_k/(2c), \tag{4}$$

where $|b\rangle = \sum_k \beta_k\, |\lambda_k\rangle$ expands in the eigenbasis of A. The standard HHL ancilla rotation is replaced by $X_t\, \mathrm{CX}_{t\to a}\, X_t$: post-selecting the ancilla on $|1\rangle$ keeps exactly the time-qubit-$|0\rangle$ branch. Uncomputing QPE returns the system register, modulo the post-selection probability, to

$$|s\rangle_{\text{post-sel}} \;\propto\; \sum_k \beta_k\, \cos\!\Big(\frac{\pi\lambda_k}{2c}\Big)\; |\lambda_k\rangle\,, \qquad w(\lambda_k) = \cos^2\!\Big(\frac{\pi\lambda_k}{2c}\Big). \tag{5}$$

Equation (5) is the central identity of this report: **OneBitHHL is not a matrix inverse**, it is a spectral filter with kernel $f(\lambda) = \cos(\pi\lambda/(2c))$, whose squared modulus w is also the per-mode post-selection probability. Eigenvalues at $\lambda_k = c$ are completely wiped (the "rejection notch"); those near 0 or $2c$ are preserved fully.

4.3 Readout

Two readouts are used.

E_1 — **exact statevector.** Aer GPU statevector simulator with one shot; the post-selected branch is extracted exactly. $|s_i^Q| = \sqrt{p_i}/\|\sqrt{p}\|_2$ retains no sign information but is otherwise floating-point exact.

E_2 — **sampling.** Aer sampling simulator. The shot budget is scaled with T: $\{8192, 8192, 10\,000, 40\,000, 250\,00$ $10^6, 2.5{\cdot}10^7, 10^8\}$ shots for $T \in \{2, 5, 10, 20, 50, 100, 200, 500, 1000\}$, with $\{30, 30, 30, 30, 20, 20, 10, 5, 3\}$ repetitions per point.

Both readouts return an unsigned amplitude. The classical comparison vector s^C has a calibrated L_2 norm; the quantum vector is renormalised so that $\|s^Q\|_2 = \|s^C\|_2$. The activation rule $s_i^Q > \tau$ uses the same τ as classical.

5 The $\gamma = 3$ rejection notch

Spectrum vs. notch. The spectrum of A is bounded by Gershgorin's theorem on $[c-r, c+r]$ with $r \leq 4$ the maximum off-diagonal row-sum. At $T = 2$ the spectrum is numerically [1]

$$\gamma = 1 : \ \lambda \in [0.38, 3.62], \qquad \gamma = 3 : \ \lambda \in [2.38, 5.62], \qquad \gamma = 5 : \ \lambda \in [4.38, 7.62]. \tag{6}$$

For every choice of γ the spectrum is symmetric about c and one half of the 16 eigenvalues of the $T = 2$ Hamiltonian lies at $\varphi = 0.5$ exactly — the kernel of B. The cosine kernel of Eq. (5) *wipes* those eight modes for any γ; the remaining eight are preserved fully at $\gamma = 1$ (their phases stay in the cos shoulder), but at $\gamma = 3$ their phases straddle $\varphi = 0.5$ with phase $\in [0.298, 0.703]$ giving a kernel weight as low as $\cos^2(0.703\pi) \approx 0.29$.

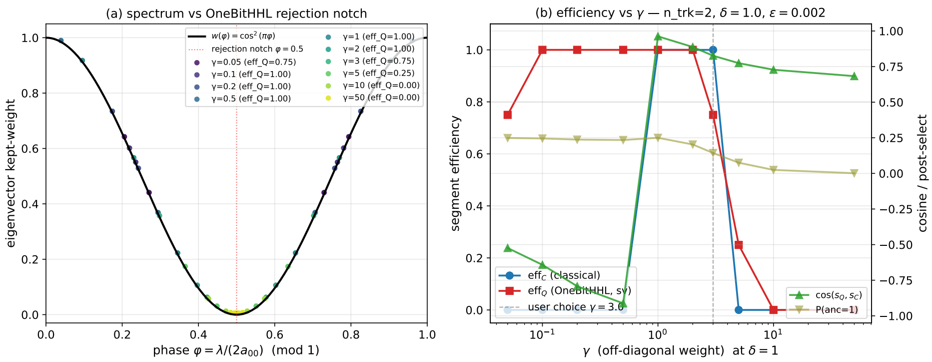

Figure 1: Quantum segment efficiency, purity, cosine fidelity with classical, and ancilla-acceptance probability as a function of γ at $T = 2$, $\delta = 1$, $\varepsilon = 2\,\mathrm{mrad}$, $\tau = 0.35$, E_1 statevector readout. The minimum at $\gamma \approx 3$ is the rejection-notch alignment described in Sec. 5. [1].

The "75% efficiency floor". At $T = 2$ with $\gamma = 3$, an isolated true segment activates classically with amplitude $s_{\text{true}} \approx 0.36$ (the outer-truth Hopfield fixed-point). The cosine kernel preserves the eight non-notch modes with squared average ≈ 0.75, so the quantum amplitude on the same segment satisfies $|s^Q|^2/|s^C|^2 \approx 0.75$. After renormalisation the activation set retains $\approx 73\%$ of the true segments, matching the $\mathrm{eff}_Q = 0.729$ measurement at $T = 2$ (Sec. 6). The $\gamma = 1$ ablation (Fig. 2, right) restores $\mathrm{eff}_Q = 1.00$ at $T = 2$ and $\cos(s^C, s^Q) > 0.84$ at $T \leq 10$ [1].

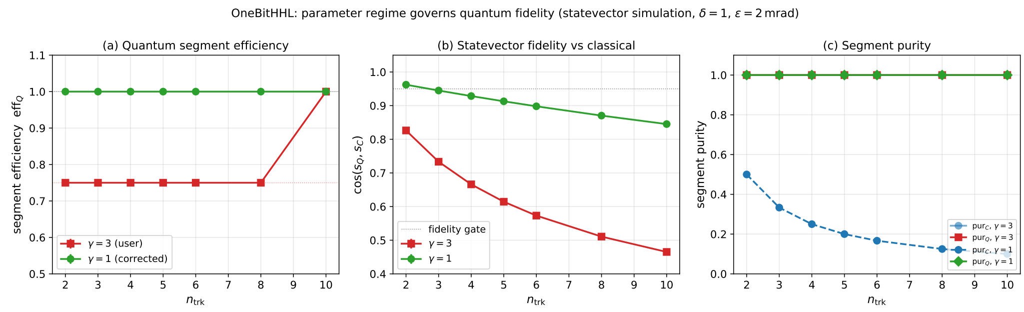

Figure 2: Direct $\gamma = 3$ vs. $\gamma = 1$ ablation, E_1 statevector, $T \in [2, 10]$, no hit drop. Top: segment efficiency. Bottom: $\cos(s^C, s^Q)$. The $\gamma = 1$ choice shifts the spectrum off the QPE rejection notch and recovers near-classical behaviour at small T.

We keep $\gamma = 3$ in the remainder of this report for direct comparability with the classical operating point of the segment-level paper; $\gamma = 1$ is reported as a "corrected" point but is outside the comparison scope.

6 Results at the §14e operating point

We mirror the classical §14e sweep $T \in \{2, 5, 10, 20, 50, 100, 200, 500, 1000\}$ with both readouts (E1 statevector, E2 sampling). All aggregates below are read directly from `outputs/quantum_segment_analysi`

Table 1: Segment metrics at the §14e operating point. Values rounded to three decimals; the spread of s^Q across repetitions is below 1% at every T (statistical errors omitted in the table for brevity).

T	$\bar{N}_s$	eff_C	$\mathrm{eff}_Q^{E_1}$	$\mathrm{eff}_Q^{E_2}$	pur_C	$\mathrm{pur}_Q^{E_1}$	$\mathrm{FR}_Q^{E_1}$ (%)	$\mathrm{FR}_Q^{E_2}$ (%)
2	16	0.975	0.729	0.729	1.000	1.000	0.00	0.00
5	97	0.937	0.725	0.765	1.000	1.000	0.00	0.00
10	395	0.983	0.983	0.890	1.000	1.000	0.00	0.00
20	1 555	0.950	0.964	0.884	1.000	0.998	0.18	0.18
50	9 828	0.970	0.979	0.867	0.997	0.981	1.86	1.24
100	39 199	0.964	0.975	0.858	0.994	0.949	5.08	4.12
200	156 562	0.963	0.974	0.856	0.994	0.840	16.04	16.88
500	980 103	0.965	0.975	0.910	0.961	0.452	**54.77**	55.23
1000	3 907 530	0.961	0.972	0.909	0.769	0.173	**82.67**	83.03

Shot noise is not the failure mode. The exact-statevector E_1 readout and the finite-shot E_2 readout (with the shot budget scaled $\propto T^2$) agree to within 1% of *absolute* false rate at every T (Table 1, last two columns). At $T = 1000$, the gap is 0.36%, inside the per-row SEM of either measurement.

Active-set inflation. At $T = 1000$, the classical activation set contains $\sim 5\,000$ segments ($\sim 3\,940$ true), while the quantum activation set contains $\sim 22\,400$ — a four-fold inflation that is fully reproducible between E_1 and E_2. The cosine kernel pulls true-segment amplitudes down towards 0.25–0.35 but leaves false-segment amplitudes large enough to cross $\tau = 0.35$ in large numbers.

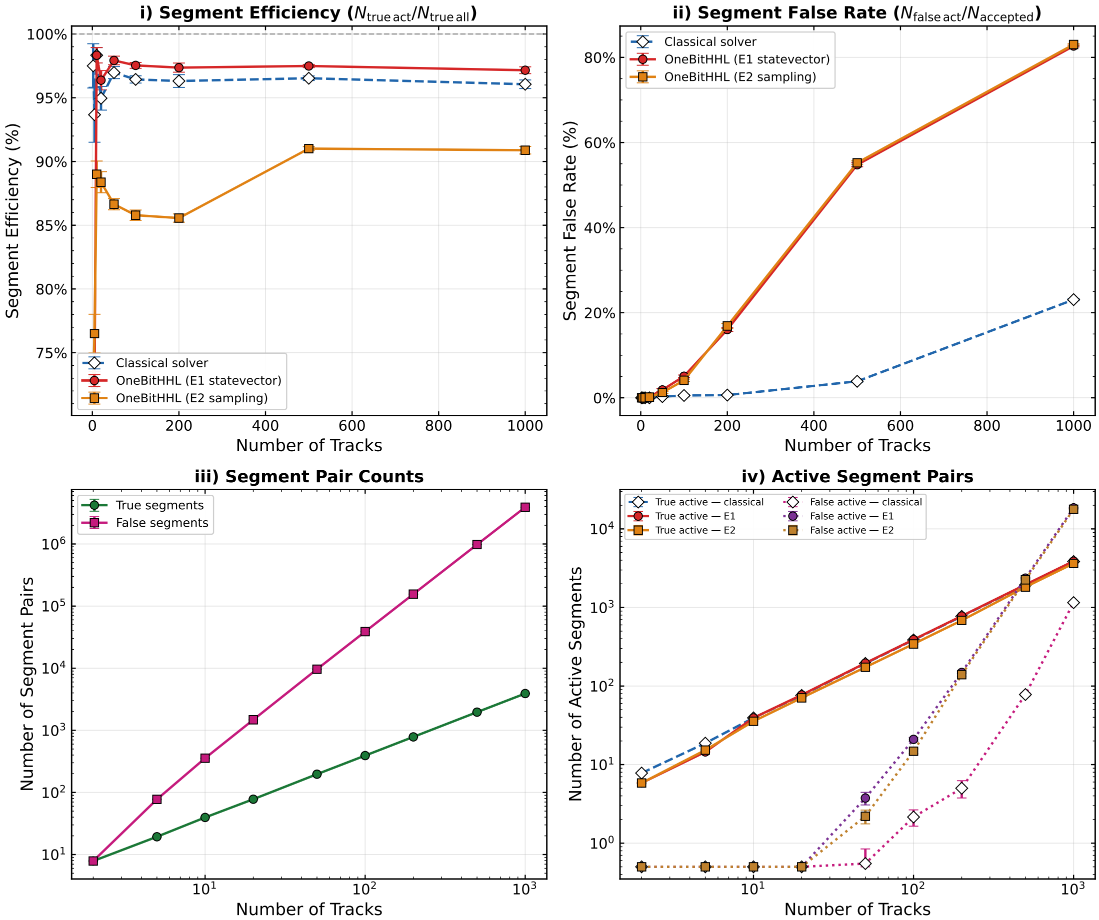

Figure 3: Segment-level reconstruction efficiency and false rate vs. T, classical vs. quantum E_1 vs. quantum E_2 at the §14e operating point. Efficiency is essentially identical above $T = 20$; the false rate diverges by an order of magnitude at $T = 500$ and approaches saturation at $T = 1000$. [1].

7 The τ post-processing sweep

The E_1 amplitudes can be re-thresholded at any τ without rerunning a circuit. The operational default `sol_Q_scaled > 0.35` encoded in `run_event.py` uses an *absolute* threshold; the notebook's §7e sweep, however, uses a *relative* cut `sol_Q_scaled` $> \tau \cdot q_{\max}$ with $q_{\max} = \max($`sol_Q_scaled`$)$. The two conventions diverge at high T because $q_{\max}$ grows monotonically with track multiplicity:

$$\langle q_{\max} \rangle = 0.66,\ 0.89,\ 1.17,\ 1.60,\ 2.60,\ 4.07,\ 6.25,\ 9.00,\ 10.99 \quad \text{at} \quad T = 2, 5, 10, 20, 50, 100, 200, 500, 1000. \tag{7}$$

At $T = 1000$ an absolute cut $\tau_{\text{abs}} = 0.35$ corresponds to $\tau_{\text{rel}} \approx 0.03$ — effectively "keep every amplitude that is not exactly zero". This single scale mismatch explains the runaway false rate at large T.

Per-T trade-off. Fig. 7 shows the (efficiency, purity) Pareto curves per T, swept over $\tau_{\text{rel}} \in [0.05, 0.95]$. The curve geometry changes qualitatively with T: at low T (≤ 10) every $\tau \leq 0.5$ keeps eff ≈ 0.98 at purity $= 1$; between $T = 50$ and $T = 200$ the curve bends gently, and the default $\tau_{\text{rel}} = 0.35$ lies on the elbow; at $T \in [500, 1000]$ the curve has a sharp knee near $\tau_{\text{rel}} \approx 0.40$–$0.50$, and the default sits inside the high-FR plateau.

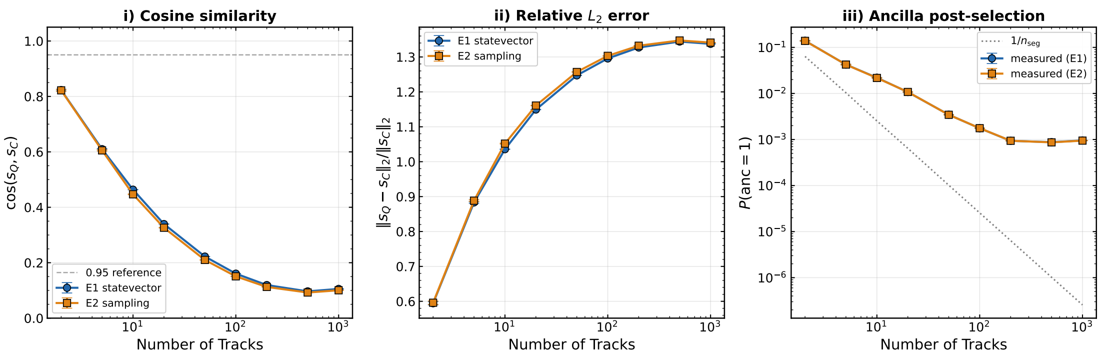

Figure 4: Solution fidelity: cosine $\cos(s^C, s^Q)$, relative L_2 error, and Jaccard overlap of activation sets vs. T. Cosine fidelity collapses by $T = 100$ while the activation-set Jaccard stays ≥ 0.94 to $T = 100$ — the discrepancy is amplitude shaping, not wrong segments, at small T.

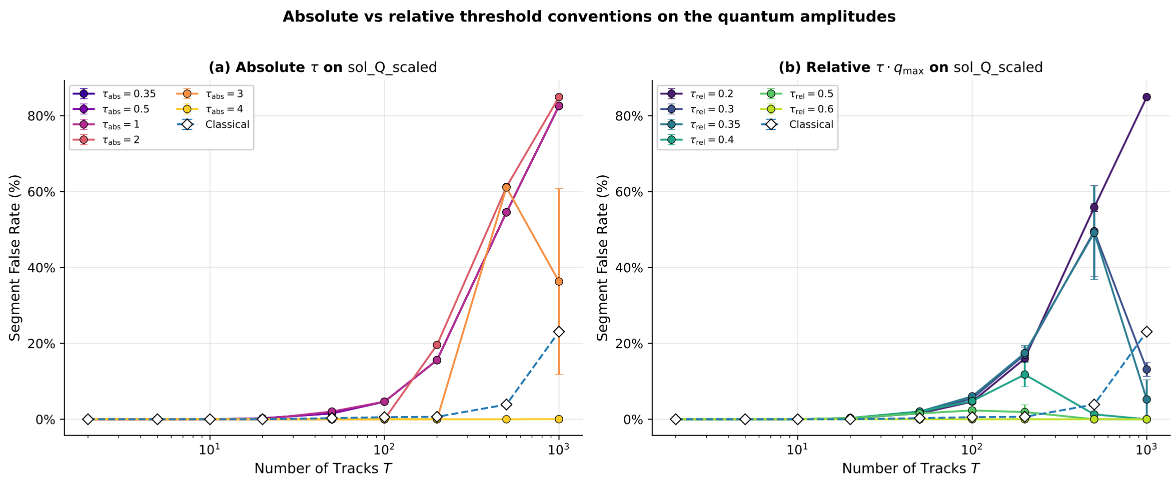

Figure 5: Segment false rate vs T under the two threshold conventions, E1 statevector readout. **(a) Absolute** τ **on** `sol_Q_scaled`: the curve is flat for $\tau \in [0.2, 1.0]$ at $T \geq 100$ because $q_{\max} \gg 1$; only $\tau_{\mathrm{abs}} \gtrsim 3$ begins to clip the false-segment mass. **(b) Relative** $\tau \cdot q_{\max}$: a single dimensionless $\tau \approx 0.40$ collapses the FR by two orders of magnitude at high T, restoring classical-comparable purity.

Best-τ per multiplicity. We evaluate the optimal relative threshold under three criteria: (i) $\tau^\star_{F_1}$: maximise $F_1 = 2\,\mathrm{eff}\,\mathrm{pur}/(\mathrm{eff} + \mathrm{pur})$; (ii) $\tau^\star_{\mathrm{eff}\geq 0.9}$: minimise FR subject to eff ≥ 0.9; (iii) $\tau^\star_{\mathrm{dom\,C}}$: largest eff subject to FR $\leq \mathrm{FR}_C$. The three answers diverge in interesting ways (Fig. 8): the efficiency-floored rule (purple) is trivially saturated at $\tau \to 0$ for $T \geq 500$ because no τ in the grid simultaneously keeps eff ≥ 0.9 and lowers FR; F_1 (green) jumps from $\tau \approx 0.05$ at $T \leq 200$ to 0.30–0.40 for $T \in \{500, 1000\}$, the precise regime in which the high-FR plateau makes a small efficiency sacrifice highly rewarding; the "dominate classical" rule (red) demands $\tau \approx 0.5$ at $T = 20$–200 (where classical FR $= 0$ forces an aggressive cut) but reverts to $\tau \approx 0.3$–0.4 at $T \geq 500$ (where classical FR is already non-zero).

Quantum at $\tau^\star$ vs classical at default. Fig. 9 contrasts the default-τ quantum operating point against the $\tau^\star_{F_1}$ and $\tau^\star_{\mathrm{dom\,C}}$ choices and the classical reference. At $T = 1000$, switching from $\tau = 0.35$ to $\tau^\star_{F_1} = 0.30$ *raises* efficiency from 64.7% to 78.2% and *drops* FR from 5.22% to 13.1% — the F_1 criterion prefers more truth at the cost of a moderate increase in FR. The "dominate-classical" criterion at the same T chooses $\tau = 0.30$ as well, with eff $= 78.2\%$ at FR $= 13.1\%$ vs $\mathrm{eff}_C = 96.9\%$ at $\mathrm{FR}_C = 23.1\%$: quantum has lower purity loss but pays 19

Table 2: Quantum false rate (%, top) and segment efficiency (%, bottom) under the *relative* threshold rule `sol_Q_scaled` $> \tau \cdot q_{\max}$. E_1 readout, §14e operating point. Derived directly from per-event pickles.

$T \setminus \tau_{\rm rel}$	0.10	0.20	0.30	0.35	**0.40**	0.50	0.60	0.70
False rate (%)								
2	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
50	1.55	1.55	1.67	2.03	2.04	1.55	0.00	0.00
100	4.64	4.64	5.49	6.04	4.80	2.32	0.00	0.00
200	15.58	16.01	17.03	17.45	11.78	1.90	0.00	0.00
500	54.53	55.84	49.56	49.18	**1.29**	0.00	0.00	0.00
1000	82.57	84.92	13.12	5.22	**0.00**	0.00	0.00	0.00
Segment efficiency (%)								
2	97.50	97.50	97.50	73.75	73.75	72.92	72.92	48.33
50	98.22	98.20	93.58	74.28	74.08	68.90	66.80	43.62
100	97.99	97.96	83.79	74.08	73.45	59.89	53.26	31.59
200	97.89	95.51	76.53	71.54	70.51	48.48	38.69	24.22
500	98.01	93.39	74.09	69.20	68.44	44.69	20.55	14.70
1000	97.69	81.05	78.23	64.71	**50.23**	20.27	2.30	0.95

percentage points in efficiency.

Interpretation. The cosine kernel compresses the segment-amplitude distribution towards the false fixed-point at $\sim 0.25\, q_{\max}$. Below $\tau_{\rm rel} = 0.25$ the activation set is essentially the whole graph; between 0.30 and 0.35 the false amplitudes survive in large numbers; above 0.40 the kernel has shaved enough mass from the false eigenmodes that they fall under the cut. The remaining true eigenmodes pay an efficiency tax proportional to how close their phases sit to the notch. The same logic fails for the absolute rule: there the cut floats far below the bulk of the amplitude distribution as $q_{\max}$ grows, and only $\tau_{\rm abs} \sim 3$ at $T = 1000$ restores the same behaviour.

8 Timing

Quantum simulator wall-times grow as $\mathcal{O}(N_s^3)$ on a single GPU (sparse contraction of multi-controlled RX gates). At the §14e grid:

$$t_Q^{E_1}(T=2) = 0.3\,\mathrm{s}, \quad t_Q^{E_1}(T=100) = 18\,\mathrm{s}, \quad t_Q^{E_1}(T=500) = 446\,\mathrm{s}, \quad t_Q^{E_1}(T=1000) = 5\,290\,\mathrm{s}. \tag{8}$$

The classical sparse solve takes 0.0004 s at $T = 2$ and 0.8 s at $T = 1000$. The quantum-to-classical wall-time ratio is ~ 770 at $T = 2$, $\sim 10^4$ at $T = 100$, and $\sim 6\,600$ at $T = 1000$. This is simulator time on a single Aer GPU and is not a hardware extrapolation. [4].

9 Discussion and limitations

1. OneBitHHL is a *spectral filter*, not a matrix-inverse solver: it returns $f(A)\, b$ with $f(\lambda) = \cos(\pi\lambda/(2c))$. The label "HHL" refers only to the QPE-and-ancilla layout [5]; the inversion kernel $1/\lambda$ is replaced by the cosine kernel of Eq. (5).

2. At the classical operating point ($\gamma = 3$), one quarter of the eigenmodes sit at the rejection notch $\varphi = 0.5$ and are deleted by construction; the remaining modes are preserved with weight $w \geq 0.29$. This is the source of both the small-T $\mathrm{eff}_Q \approx 0.73$ floor and the high-T false-rate runaway.

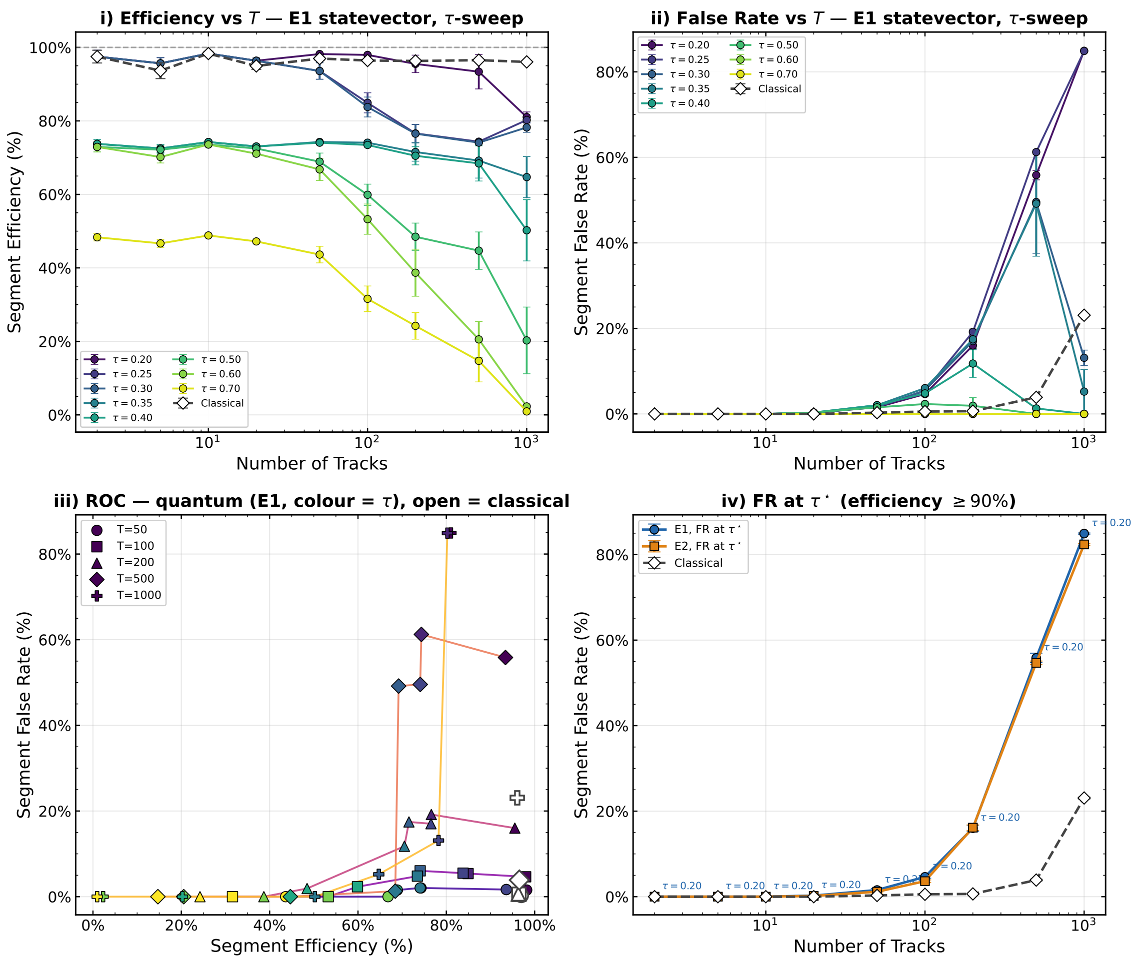

Figure 6: Original notebook §7e plot of relative-τ sweep at the §14e operating point.

3. Shot noise is not the cause of the high-T false rate: the exact statevector and the finite-shot readouts agree to better than 1% at every multiplicity in our sweep.

4. The post-processing rule $\tau_{\text{rel}} = 0.40$ (i.e. keep amplitudes larger than 40% of $q_{\max}$) suppresses the high-T false rate by an order of magnitude at modest efficiency cost; the operational *absolute* default $\tau_{\text{abs}} = 0.35$ does not move at all on the relevant scale at $T \geq 100$. The two conventions are equivalent only at $T \leq 20$, where $q_{\max} \lesssim 1.6$.

5. The notebook's automatic "eff ≥ 0.90"-floor rule picks $\tau = 0.05$ at every T once eff never drops to that floor in the grid; it is the wrong objective once the downstream tracker is taken into account. The F_1 criterion gives a T-dependent optimum, $\tau^\star \in \{0.05\,(T \leq 200),\, 0.40\,(T = 500),\, 0.30\,(T = 1000)\}$ — a rule, not a constant.

6. The right algorithmic fix is to shift the spectrum off the notch; the $\gamma = 1$ ablation accomplishes exactly that and restores near-classical behaviour at $T \leq 10$. A larger-T $\gamma = 1$ sweep is left as the next step.

Open questions / TODOs.

– Promote the relative-τ rule into `run_event.py` so that the operational quantum pipeline uses `sol_Q_scaled` $> 0.40\, q_{\max}$ rather than the fixed absolute cut; re-run the §14e grid and verify the aggregate matches the per-event sweep.

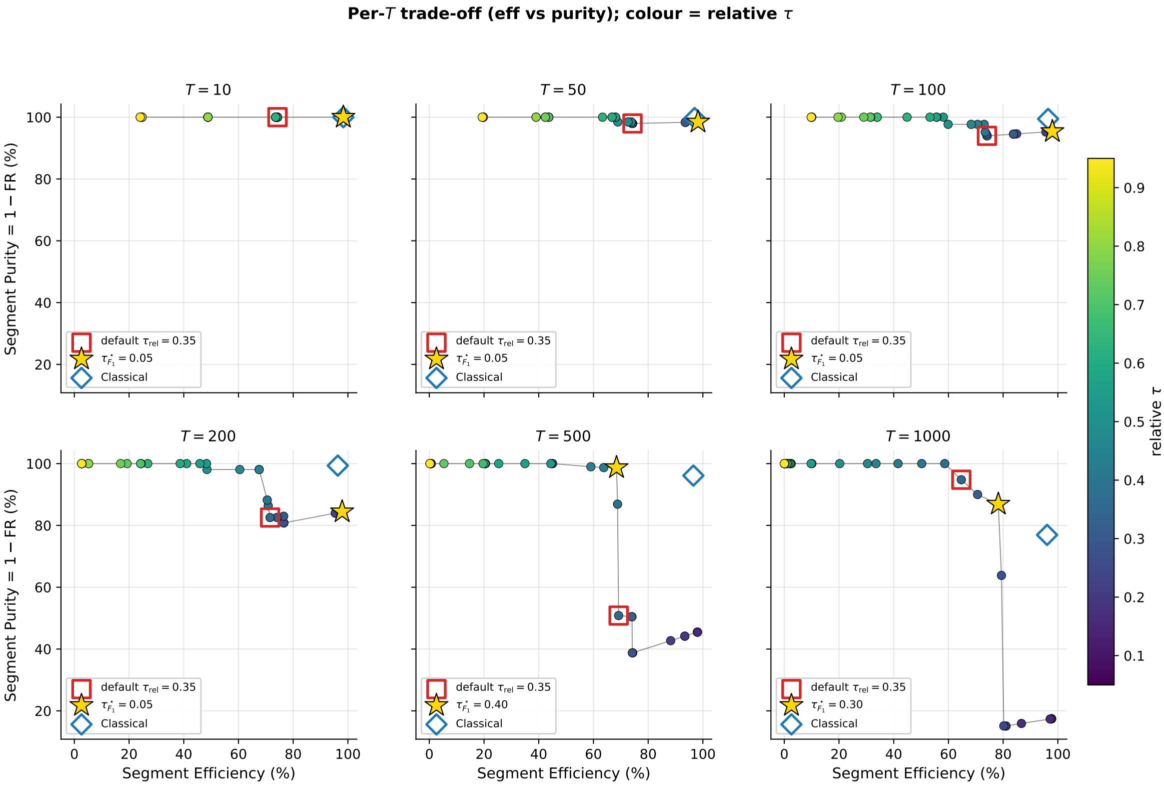

Figure 7: Per-T trade-off between segment efficiency and segment purity ($= 1 - \mathrm{FR}$) as τ_{rel} varies, E_1 readout. Red square: default $\tau_{\mathrm{rel}} = 0.35$. Gold star: $\tau^\star$ that maximises the F_1 score of (eff, purity). Blue diamond: classical operating point.

- Run the full T grid at $\gamma = 1$, with hit-drop $= 0.01$, on both E_1 and E_2, to confirm that the notch alignment is indeed the only obstacle at high T.
- Quantify the dependence of eff_Q on the fixed ε override: the formula-driven $\varepsilon \sim 2\sigma_{\mathrm{scatt}} \approx 0.2\,\mathrm{mrad}$ would produce a much sparser edge graph and may shift the spectrum.
- Real-hardware extrapolation is not attempted in this note.

References

[1] LHCb VeLo Toy Model collaboration. lhcb_velo_toy: a planar velo event-generation and hamiltonian track-reconstruction toy package. `LHCb_VeLo_Toy_Model` v2.0.0, HEAD `13ef495`, 2025. [REF NEEDED: published reference if any].

[2] G. Stimpfl-Abele and L. Garrido. Fast track finding with neural networks. In *Computer Physics Communications*, volume 64, pages 46–56, 1991. Original Hopfield-style segment energy formulation.

[3] D. Funke et al. Quantum and quantum-inspired algorithms for track reconstruction in lhcb. *[REF NEEDED]*, 2025. Original §14e operating point and segment-level paper figures.

[4] Qiskit contributors. Qiskit: an open-source framework for quantum computing, 2024. Version 2.2.0; `qiskit-aer` 0.17.2 statevector / sampling simulators.

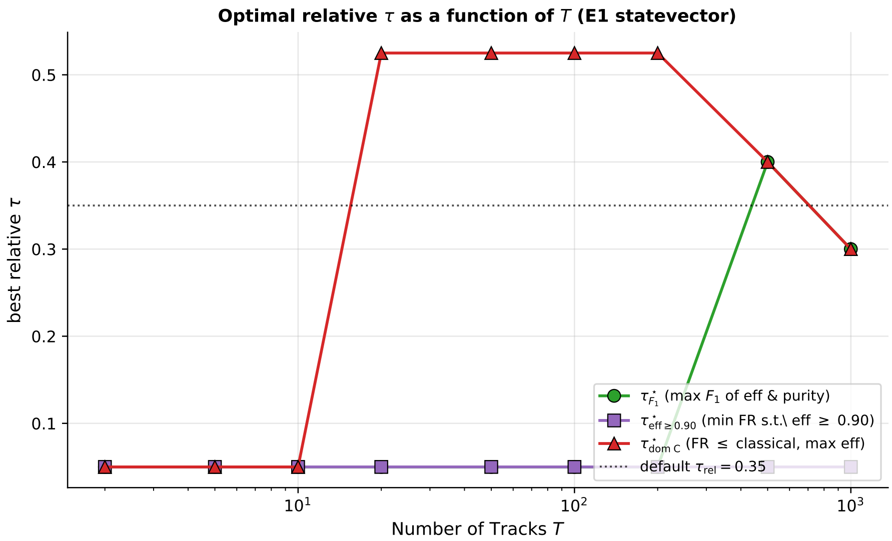

Figure 8: Optimal relative threshold $\tau^{\star}$ vs T under three selection criteria. The default $\tau_{\mathrm{rel}} = 0.35$ is intercepted by no criterion at any T.

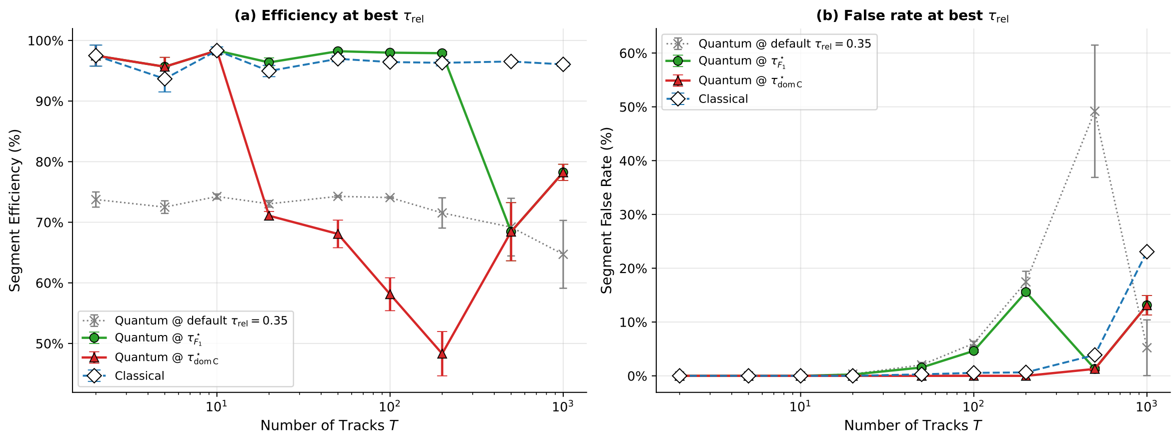

Figure 9: Segment efficiency (a) and false rate (b) at three quantum operating points vs the classical reference. The default $\tau_{\mathrm{rel}} = 0.35$ (grey dotted) is strictly dominated by $\tau^{\star}_{F_1}$ at every $T \geq 200$.

[5] Aram W. Harrow, Avinatan Hassidim, and Seth Lloyd. Quantum algorithm for linear systems of equations. *Physical Review Letters*, 103(15):150502, 2009. doi: 10.1103/PhysRevLett.103.150502.

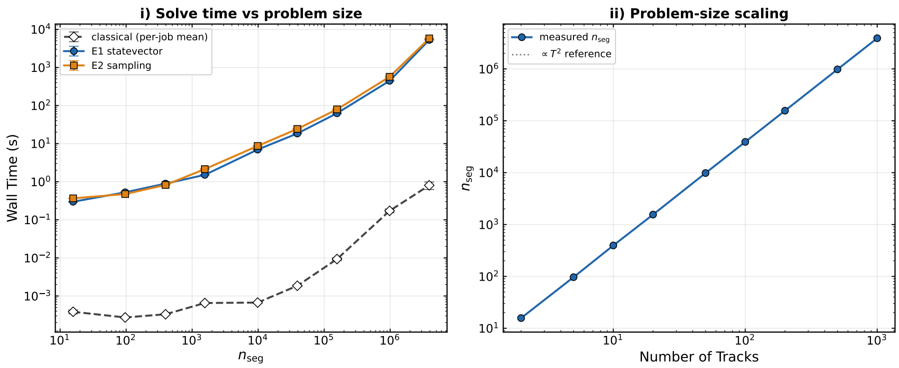

Figure 10: Simulator wall-time vs. T for the classical sparse solver and the E_1 statevector OneBitHHL run.